

Receiver psychology and the evolution of animal signals

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Abstract. Despite decades of interest, the adaptive significance of the extraordinary diversity in the design of animal signals remains elusive. It is suggested that signal design consists of two components: 'strategic design' and 'efficacy'. Strategic design is concerned with how a signal is constructed by natural selection to provide the information necessary to make a receiver respond (e.g. by being good at displaying underlying quality), whilst efficacy is concerned with how a signal is designed to get that information across to the receiver (e.g. by being easily measured). It is argued that an important but neglected evolutionary force on animal signals is therefore the psychology of the signal receiver, and that three aspects of receiver psychology (what a receiver finds easy to detect, easy to discriminate and easy to remember) constitute powerful selective forces in signal design. Greatest emphasis is given to memorability because this has been least considered by previous authors. It is argued that learning and memory are involved in a wide range of signals, and numerous hypotheses as to how signals may be adapted to be more memorable to receivers are suggested. The relationship of this analysis to earlier attempts at understanding signals is explored, particularly with reference to the concepts of honesty, manipulation and mind-reading.

From the time of Darwin (1871), through the classic work of Huxley (1914) and Tinbergen (1959), animal communication has been the object of intense interest to biologists. Yet, despite a great deal of study, genuine mystery still surrounds the evolution of animal signals. Thumb through any bird guide, walk through a tropical forest full of butterflies, swim over a coral reef busy with fish and it will be clear that none of our existing theories explains the sheer diversity of different colours, sounds and behaviour that constitute 'signals' in different species.

We believe that the enormous variety in the design of particular signals is the product, at least in part, of a rich variety of selective forces deriving from the psychology of the intended receivers of those signals. In this paper, we wish to draw attention to the largely unexplored problem of how receiver psychology influences the evolution of signals. We argue that three features of the receivers' 'psychological landscape' (detectability, discriminability and memorability) contribute to the design of all animal signals and go a long way towards explaining why signals are so different in different species. Of these three features, we give particular prominence to memorability because we believe that this influence on signal design has been least emphasized by other authors. We show that many signals

depend for their effect on the ability of receivers to learn and to remember and that, consequently, design for memorability (or in some cases lack of it) has been a powerful evolutionary force determining why some signals are the way they are.

The role of receiver psychology in animal communication is best understood by realizing that signal design has two distinct components. The first, which we call 'strategic design', concerns the evolutionary forces acting on a signaller to give a particular type of signal, and on the receiver to respond to that signal. Strategic design might, for example, arise from the evolutionary pressures on females to select healthy, disease-free mates (Hamilton & Zuk 1982) and those on males to display their healthy, disease-free state. But while many species might have the same overall evolutionary strategy of choosing disease-free mates, the particular signal one species employs to achieve this may be quite different from the signal another species uses. To understand the selective pressures operating on particular signals, then, we need to consider a second component of design: the 'tactical design' or 'efficacy' of signals. It might be, for instance, that in one species females are attracted to bright red throat patches and in another to long black tails. The difference might arise from what makes an effective signal in their

two environments (e.g. Morton 1975; Wiley & Richards 1978; Hunter & Krebs 1979) or, as we argue, what is eye-catching, easily discriminable or memorable to the brains of their respective females.

Another way of expressing this difference is to say that the first, strategic, component of signal design is what determines whether or why (in fitness terms) the signal receiver responds appropriately, whereas the second, efficacy, component affects the probability that the signal, once given, will reach its target destination and elicit a response at all. Efficacy can be understood only by considering the details of the environment, sense organs and brain of the receiver.

Zahavi (1987) implicitly recognized the distinction between these two components of signal design in the evolution of threat signals. He wrote 'A movement sideways may be easier to observe and therefore clearer than a movement forward. However, out of all possible directions a forward movement towards the rival has universally been chosen to signal high motivation to attack. I suggest that the risk involved in moving towards the rival has selected the forward movement as the optimal threat signal because it is the most reliable one.' This view is, however, one-sided. Reliability is only part of the design of the signal. Out of all the possible reliable signals that animals can use, the one that will be chosen (in an evolutionary sense) will be the one that is most effective. Even the most honest signals have to get their message over to the receiver.

It is therefore the second, efficacy, component of signal design that we emphasize in this paper. However, since the strategic component is also an important determinant of what a signal is like, we discuss this and the controversies of whether signals are honest (Zahavi 1987; Grafen 1990b) or manipulative (Krebs & Dawkins 1984) in a later section. We do not deny that strategic considerations and the messages that signals carry affect the design of those signals. What we do wish to do is to point out that strategic considerations are not enough to explain differences between species in the design of their signals. There is more to signals than their messages: we must also consider their efficacy as carriers of messages.

THE PSYCHOLOGICAL LANDSCAPE

Obviously signals must be transmitted effectively from one organism to another if they are to serve

their function of altering the behaviour of that other organism. There have already been many attempts to understand the design of signals in terms of the efficiency of transmission through the environment (Konishi 1970; Hailman 1977; Smith 1977; Gerhardt 1983; Endler, *in press a*). Studies relating the nature of birds' songs to the habitat in which they are used are a good example (Morton 1975; Hunter & Krebs 1979; Wiley & Richards 1983). Signal evolution has also been interpreted in terms of adaptation to the properties of sensory systems (Endler & McLellan 1988; Pomiankowski & Guilford 1990; Ryan & Rand 1990; Ryan *et al.* 1990; Endler, *in press a*), and the constraints imposed by competing selection pressures (e.g. Endler 1978; Baker & Parker 1979). But in considering the effectiveness or efficacy of a signal, we have to look beyond the medium through which a signal is travelling and even beyond the sense organs of the receiver animal to what we call the receiver's 'psychological landscape'. By 'psychological landscape' we mean everything about the brain of the receiver animal that might affect its response to a signal. Suppose that a receiver finds certain features of signals attention-getting or memorable, for reasons quite unconnected with signalling, then these features are likely to become incorporated into the design of some signals (and avoided in the design of others) just as surely as those features that minimize the distorting or attenuating effects of the medium through which the signal has to travel. For instance, if a female bird is attracted to movement, symmetry, or red colouring, when she searches for food, then male birds of that species might be expected to make use of these predispositions and exploit them in their courtship signals.

With the exception of Dawkins & Krebs (1978; Krebs & Dawkins 1984), most analyses of signal design seem to have stopped at the level of the eye or the ear, as if that were the final destination of the signal. But up behind the sense organs, of even relatively simple organisms, is a bewilderingly complex system of processors, information-storers and decision-makers. The brain of any animal has its own history of ancestral innovations, changes in capacity, divisions of labour and efficiency compromises, all leading to a system with its own set of distinctive characteristics. Some characteristics, such as sensitivity to movement, may be more or less universal, but others may be idiosyncratic and specialized. The brain provides not a featureless black box, but a complex psychological landscape

across which messages must pass and to which signals have had to evolve.

Consider one important feature of a signal's efficacy: its detectability (how easily it can be perceived as distinct from its background). Detectability is not just the property of the medium through which the signal travels, or of the sense organs of the receiver, but also of the brain of the receiver itself. And detectability itself is only one part of the psychological landscape. Two other features which may be just as important are a signal's 'discriminability' (how easily it can be separated from other stimuli with which it may be confused) and 'memorability' (how easily it can be remembered, or associated with some other relevant stimulus). Detectability, discriminability and memorability clearly interact. Aspects of a signal that lead to its being increasingly detectable may also lead to its being increasingly discriminable, whilst how memorable a signal is may in turn be affected by how discriminable that signal is from other signals. Nevertheless, we consider that it is helpful to draw a distinction between these three features because they cover a range of important operations that the brain performs on incoming sensory information, and therefore to which the signals of another animal will have evolved.

Detectability

Detectability has been well considered before as a property of signals (Hailman 1977; Wiley 1983; Wiley & Richards 1983; Fleishman 1988). One part of what makes a signal detectable involves parameters that require little knowledge about the psychological landscape. These are physical parameters such as intensity, duration, repetition rate, or size. Larger, and more intense signals will be detectable from a greater distance, and longer or more repetitive signals have more chance of being detected by a receiver that is only intermittently attentive (see also Wiley 1983). At least some of the diversity in the design of signals can be understood by analysing how the physical characteristics of different environments, of different media, and of different sense organs, should affect transmission, and therefore detectability (Levine & MacNichol 1979; Lythgoe 1979; Endler & McLellan 1988; Ryan et al. 1990). Birds living in forest environments tend to have lower frequency songs than those in open habitats (Morton 1975), even within

the same species (e.g. *Parus major*: Hunter & Krebs 1979), probably because lower frequencies suffer less degradation from echoes off objects such as leaves than do high frequencies (Wiley & Richards 1978). To take a visual example, male guppies, *Poecilia reticulata*, display to females using spot colours that are designed to be more conspicuous to female guppies than to their dangerous predator *Crenicichla alta* by exploiting differences in the viewing conditions and spectral sensitivities of the two receivers (Endler, in press a). Plants use ultraviolet signals to be detectable to insect pollinators (Eisner et al. 1969), or perhaps birds (Chen et al. 1984; Chen & Goldsmith 1986; Burkhardt 1989; Maier & Maier 1989), but cannot do so to attract mammals that are insensitive to ultraviolet. Colour contrast with the background may enhance detectability, but which colours are most effective will depend on the colour sensitivities of potential receivers (e.g. Lall et al. 1980).

However, it is not just the transmission environment and the sense organs that determine the detectability of a signal. The psychology of the receiver will be critical too. For example, visual perception seems to work by building the responses of specialized feature detectors into recognizable objects (e.g. Frisby 1979; Roth & Frisby 1986; Treisman 1986), and receivers may be better at detecting some features than others. Humans are better at detecting vertical and horizontal lines than oblique ones (Frisby 1979). Amphibia have feature detectors specialized for detecting prey-like shapes (e.g. Ewert 1984; Roth 1986). Receivers may also be specialized in the detection of movement (e.g. Ewert & Hock 1972), perhaps because it signals danger, or food. Many displays seem to start with a vigorous movement which may serve to draw the receiver's gaze by exploiting this fact (the head-bobs of lizards, for example: Fleishman 1988). Once a receiver's gaze or attention has been shifted towards the emitter, this may allow the receiver to concentrate more effectively on the rest of the display, or to enhance detectability of subsequent stimuli by 'alerting' it (Richards 1981), perhaps through encouraging selective attention. We suggest that drawing the immediate attention of receivers may also be one function of song-matching in the territorial disputes of song birds (Krebs et al. 1981; Falls et al. 1982) in a way analogous to the 'cocktail-party' effect in humans (Cherry 1953). In a room full of conversations, human receivers are

capable of attending specifically to a target conversation, remaining apparently oblivious of the content of others (though they do apparently process much more information than they are aware of: Broadbent 1982; Johnston & Dark 1986). However, if their own name (or any one of a limited set of words which carry special salience to the receiver) is uttered in one of these other conversations, they notice it immediately, and shift attention to its source (Treisman 1960, 1964). Perhaps, by singing a song closely matching the one they have just heard, the territory holder increases the detectability of his own song to the intruder literally by uttering the intruder's name amidst an otherwise potentially confusing cacophony of background sounds.

Drawing the attention of receivers may also be the function of some mimetic stimuli. We suggest, for example, that the use of egg-mimicking spots on the anal fin of mouth-brooding cichlids may help in rapid assessment of male body size by causing receivers to focus their attention on a part of the body that is a good indicator of size (rather than enhancing fertilization in line with Wickler's (1962) hypothesis which is not supported by Hert's (1989) experimental evidence). Perhaps the use of prominent eye spots in the tail of the peacock, *Pavo cristatus*, may facilitate the counting of tail feathers, inexorably drawing the female's gaze because she cannot normally afford to ignore eye-like objects. In these cases the use of subliminal associations (eyes or egg-spots, that are already powerful stimuli in another context) may enhance the receiver's attention to a male's relevant features. Certainly, where eye-spots are used defensively, as in the peacock butterfly, *Inachris io*, prey are keying into some powerful response in potential predators (Blest 1957; Scaife 1976; Jones 1980).

Rothschild (1964a, 1985) has suggested that some warning colour patterns may be cryptic at a distance, but conspicuous close to. If the spatial frequencies of spot or stripe patterns were designed to offer high contrast boundaries within certain detection ranges, yet also designed to blend to an inconspicuous whole beyond this distance, then we would see a diversity of patterns depending on the detection ranges of different receivers. On the other hand, conspicuous warning patterns may be designed to be detected when there is still a considerable distance between predator and prey so as to increase the probability of correcting recognition errors in experienced predators before capture

(Guilford 1986). On this hypothesis, the prey's signal should be designed to be as conspicuous as possible to the brain of the predator. Many prey animals, of course, do not benefit at all from being detected, and their colour patterns and behaviour may be understood in terms of strategies to evade capture by predators that have already learnt to detect cryptic prey (Guilford & Dawkins 1987; Allen 1988; Endler 1988).

Signals involved in the assessment of qualities relevant to fighting ability or resource holding potential are also designed for detectability. In other words, as well as the strategic component of their design (quality advertisement), there is an important efficacy component as well. This can be illustrated by the classic case of roaring in male red deer, *Cervus elaphus*. Before or instead of a fight, stags will enter a roaring contest which, if unresolved, is followed by a series of parallel walks. Clutton-Brock & Albon (1979) argued that roaring uses the same thoracic muscles as fighting and that, since roaring is tiring, an ability to maintain a high roaring rate is a good indicator of ability to fight hard without tiring. If, further, roaring is physiologically more costly to weak individuals than strong ones, then it could honestly advertise fighting ability in line with the handicap principle (Zahavi 1987; Burk 1988; Grafen 1990b). It is interesting, however, that even in this relatively straightforward example of quality advertisement, the strategic component is not the only one we have to consider. Although roaring may contain accurate information about fighting ability (with honesty of the signal maintained by cost) it is unlikely to be the only activity that could do so. A stag might be able to tense its fighting muscles directly, and the duration over which it could do so might accurately reflect fighting ability. But this activity would be no good in quality advertisement because it would be difficult to detect, so would be effort wasted. It is surely no coincidence that of all the potential correlates of quality, the one used in advertising is highly detectable.

Sexual displays are also often designed for detectability. Male birds often sing from prominent perches or display on leks where the presence of other males will already be attracting a large female audience. Male frogs (Ryan et al. 1990) and guppies (Endler, in press a) use display frequencies suited to biases in female sensory sensitivity. Fruit and flowers are also designed to be detectable from a distance (Marler 1977; Willson & Mellampy 1983).

Discriminability

Discrimination involves recognition in the signal receiver that a stimulus, or configuration of stimuli, belongs to some discrete category. The importance of design for discriminability has been recognized since Wallace (1867) suggested that distasteful insect larvae '... required some distinctive mark, something by which they may be contrasted with and separated from the agreeable larvae, in order that they might be freed from the attacks of birds' and that 'Brilliant colouration would be such a distinction as was required' (page lxxx). Warning colours and patterns should therefore look different from those of the prey for which predators normally hunt (Darwin 1871; Poulton 1890; Edmunds 1974). Most palatable prey present patterns that are undetectable against their normal backgrounds (Endler 1978, 1984), whereas warning signals consist of intense colours, particularly reds, yellows, oranges and even whites that make discrimination from palatable prey particularly easy. Warning colour patterns may also increase discriminability from palatable prey by enhancing regularity through the use of repeating designs such as spots and stripes. By presenting predators with a consistent signal that looks similar from all angles of attack or even if the prey is partially obscured by foliage (Guilford 1990), the distasteful insect makes it even less likely that it will be confused with palatable prey.

Discriminability is important in quality advertisement signals too. The roaring of red deer is a signal that can be given at different levels, and it is the level that reflects quality such as size or strength. Efficacious quality advertisement signals should make it easy for the receiver to discriminate different levels if it is to the advantage of the signaller to be seen as clearly distinct from other signallers by exploiting receiver sensitivity (Krebs & Dawkins 1984; Ryan & Rand 1990; Ryan et al. 1990). Design for discriminability may be predicted from a knowledge of the sensory sensitivity of receivers, but here too the psychological landscape is important. Suppose that the duration of a display is a good correlate of quality (Burk 1988). Animals appear to measure and remember the duration of signals lasting a few seconds more accurately the shorter they are (e.g. Church & Gibbon 1982). Hence receivers might be expected to gather more about a signaller's quality from a short but intense display than from a longer, less intense type of display, even if the duration of both were equally good correlates

of quality. The possibility that the addition of other features, such as rhythmicity or repetition, might also affect the accuracy of memory for duration does not seem to have been investigated, but could clearly be important in signal design.

In other ways, too, the psychological landscape of the intended receiver can affect the type of signal used. Zahavi (1987) has suggested that the length of a fish may be more easily measured if accentuated by a horizontal line. The widespread occurrence of stripes could therefore be due to design for size discriminability, where it is important for animals to signal their true size. Hasson (1989) also recognized that some signals may function to improve receiver discrimination power. He termed such signals 'amplifiers', explicitly distinguished them from the 'quality indicators' they are designed to amplify, and showed that they can evolve in genetic models. Hasson's amplifiers seem to be concerned with signal efficacy because they are designed to make the discrimination of quality advertisement levels easier.

On the other hand, where it is of advantage for animals to conceal their size or give the impression that they are larger than they really are, lines or other devices may be used deceptively. We know that pattern characteristics certainly influence the perceived size and shape of objects for humans and we call such phenomena 'visual illusions' (Gregory 1977; Frisby 1979; Roth & Frisby 1986). For example, artificial distance cues (such as perspective) can make things look bigger than they are (the Ponzo illusion), and vertical lines usually look longer than horizontal ones. Revesz (1924) and Winslow (1933) found that chickens, *Gallus gallus domesticus*, appear to see some of the same visual illusions that humans do, for example, apparently seeing lines as longer or shorter than they really are in particular settings. If this is a widespread phenomenon among other animals, many signals may be designed to exploit this fact. By judicious use of spots or lines, animals could make themselves, or parts of themselves, look bigger or smaller, fatter or thinner than they really are. Even in perception of lightness and colour, the brain distorts incoming sensory data to construct its own illusory version of the world outside. The perceived brightness of patches and their colours is not just a one-to-one translation of their physical intensity and wavelength, but is influenced by the sharpness of edges and the colours of neighbouring patches (e.g. Frisby 1979; Brou et al. 1986). In other words,

patches of colour look more vivid if they have a sharp boundary with another colour than if the two colours grade into one another. Could this be why warning colours so often involve stark colour boundaries such as stripes? If so, then we might predict it to be true for some other signals such as sexual displays as well.

Another phenomenon that may be of considerable importance in signal design for efficacy is known as 'peak shift' which is a well documented, general effect in discrimination learning (e.g. Hanson 1959; Staddon 1975; Rilling 1977; Tarpay 1982). Discrimination is normally achieved by the tension between the excitatory effect of likeness to a desired object, and the inhibitory effect of likeness to non-desired objects. Where discrimination is between two quite similar stimuli in the same stimulus dimension, peak responding occurs not at the exact position of the positive stimulus, but beyond it in the direction away from the negative stimulus: the peak is shifted. It is as if animals are even more sure that a stimulus is positive because it is further away from the negative stimulus than the originally positive stimulus. Peak shift may be important in signalling because it suggests that the most effective way to appear to belong to a particular desired category is to display exaggerated characteristics rather than average ones for the category. If females learn to discriminate males from other females (e.g. through sexual imprinting), peak shift might cause preferences for exaggerated males thus providing a mechanism for sexual selection (D. Weary, T. Guilford & R. Weisman, unpublished). If females confuse males of two closely related species, then this might set up apparent preferences for exaggerated signals from males of their own species, leading to rapid divergence in secondary sexual characters, at least until the two species are sufficiently distinct that there is no possibility of confusion to encourage peak shift.

Peak shift occurs where a difficult discrimination has to be learnt (see next section), but signallers may also benefit from exaggerated displays where no learning is involved, but where there is a hard-wired, 'supernormal' response (Tinbergen 1948; Magnus 1958). Peak shift and supernormal responses both arise from the need to discriminate positive from similar negative stimuli (Hogan et al. 1975; Staddon 1975) and both suggest reasons why exaggerated signals can sometimes be the most effective way of producing a response in a receiver.

Memorability

The role of learning

At first sight, it may not be obvious that memorability is a design feature of signals at all. But wherever learning is involved in communication, memory will be important. In the following sections we first establish the relevance of learning and memory in both inter- and intraspecific communication, and then explore ways in which memorability considerations can mould signal design.

The importance of learning is perhaps most obvious in interspecific defensive signalling. Warning colours, for example, usually derive their impact from the fact that predators learn the association between unpalatability and appearance (reviewed in Guilford 1990). The predator gains from speeding up the learning process, and decreasing the number of unpalatable prey it samples and so do the prey, especially when the learning process is likely to be dangerous to them as individuals (with slow acting defensive toxins, for example). Ap Rhisiart (1989) has shown that adult male skylarks, *Alauda arvensis*, sing at approaching merlins, *Falco columbarius*, to indicate that they belong to a category of prey that are unlikely to be caught, a form of 'pursuit deterrence'. Inexperienced merlins chase adult male skylarks, and only understand the signal once they have learnt its meaning. Pursuit deterrence song, and other similar signals (e.g. Bildstein 1983), should be designed to be memorable.

Communication within species, too, may involve considerable components of learning and memory. Social dominance hierarchies in a variety of species seem to rest on learnt recognition (e.g. Wade 1978; Shawcross & Slater 1984; Nilsson 1989), although not necessarily of individuals (Barnard & Burk 1979). Animals often learn their own status relative to others, that is, they calibrate their own strength or other indications of quality. Experience, particularly that of winning or losing fights, is known to be important in many instances (Ratner 1961; Beacham & Newman 1987).

Even in intraspecific assessment signalling (Clutton-Brock & Albon 1979; Maynard Smith & Harper 1988) an individual may benefit from having a signal at least partially because of what receivers take it to mean from their past experience with other signallers. As with warning colours, receivers will use their memory of signals which in

the past have been associated with quality (such as fighting ability) and generalize to other individuals having the same signal, rather than fully assess them at each encounter. The plumage 'badges of status' found in many bird species (Roper 1986; Whitfield 1987; Maynard Smith & Harper 1988) may be interpreted in this light. Plumage badges do not appear to be necessarily correlated with fighting ability; rather they are conventional signals open to low levels of cheating (Maynard Smith & Harper 1988), with some degree of honesty maintained by the fact that cheats with higher status badges than their true quality warrants suffer costs of increased attacks by true dominants (Rowher 1975; Møller 1987). Because constant full assessment would be costly, receivers rely at least partially on what they have already learnt about the meaning of each badge. These arguments are developed in more detail elsewhere (Dawkins & Guilford 1991) but the important point here is that if any part of a signal's effect is derived from its similarity to something the receiver has already learnt, then this will in turn have a major effect on the design of the signal that is used: the signal should be memorable and distinctive.

When females choose males as mates, learning may also be important, particularly if the female 'shops around' or remembers something about one male in relation to another. Even a simple rule of female choice such as 'choose the largest male encountered in a given time period' involves memory. In fact, females of many species choose males not on the basis of the properties of the male himself, but on the basis of attributes such as territory quality (e.g. Alatalo et al. 1986). In such cases, the female has to remember the association between a particular male and a location, and then return to it. If the female is choosing males not on the basis of territory or resources but for genetic quality, learning may also be involved for other reasons. If female choice involves visiting (and assessing) several males in turn before mating, as appears to be the case in peacocks (M. Petrie, personal communication), females will have to remember what males are like. They will have to associate a male's quality with his identity and/or his location and return to the best one later. Losey et al. (1986) argued that females may avoid some of the costs of assessing males by copying the choices of other females, but they will still need to remember which males have been chosen, and, perhaps, by whom.

What makes a signal memorable?

Having seen that learning and memory can be important in a diversity of signalling contexts, we can ask: what makes a signal easy to learn and remember? We know that animals can learn connections between events in their world, such as the palatability of food with a particular appearance, or the painful consequences of touching certain objects. However, we also know that such connections may be more easily made, or retained, between some pairs of events than between others (e.g. Garcia & Koelling 1966; Hinde & Stevenson-Hinde 1973; Shettleworth 1984). All sorts of factors determine how easily a connection is made in the first place, and for how long that connection will be retained in the brain. These include the nature of the events themselves (e.g. whether there is a 'reward' or 'punishment'), how intense or familiar they are, how consistently the events are related, the animal's own motivational and attentional state, the presence or absence of prior associations, the presence or absence of distracting or potentiating stimuli, and so on (Mackintosh 1974, 1983; Tarpy 1982). We may not yet know what all these rules are, but if we want to know what makes signals memorable then we will need to find out.

The rules of memorability in animal communication may be quite different from those of the food-related tasks that have most often been studied (Mackintosh 1983). What makes a signal memorable in location learning may be quite different when a high quality mate rather than food is the reward. Nevertheless, some principles must govern learning in these unusual contexts, and the facts about animal learning that have been discovered so far provide a starting point.

Contrast. Gittleman & Harvey (1980) were perhaps the first to point out the importance of memorability in the design of warning colours. They showed that chicks apparently learn to avoid unpalatable prey faster if these are presented so as to contrast with the background (see also Roper & Wistow 1986). Roper & Redston (1987) subsequently demonstrated that even after a single trial contrast increases the durability of aversion learning. Since these chicks were naive, the implication is that their brains are constructed in such a way that associations between unpalatability and coloration are more readily made, and last longer, if the

coloration contrasts with the background. If contrasting signals are especially memorable, then we might expect the evolution not just of colours bright against the background, but also of patterns containing internal high contrast boundaries (Guilford 1990). The bicoloured fruit displays of many plants (Willson & Melampy 1983) suggest that they too might be benefitting from exploiting similar memorability biases, this time in the minds of dispersers learning positive associations with food rewards. The effects of bright alluring colours of flowers on pollinators could be similar. The role of memorability through contrast in other areas, such as status signalling or sexual signalling, remains to be investigated.

Specific colours and patterns. Whilst some experiments clearly demonstrate that memorability is caused by contrast, and is independent of colour itself, Schuler & Hesse (1985) suggested that black and yellow bands make effective warning patterns because predators may be predisposed to learn to avoid them. Certainly, animals do seem to have prior expectations about the value of different colours or patterns (Caldwell & Rubinoff 1983; Roper & Cook 1989; Roper 1990). Perhaps, then, there is also something especially memorable about particular warning colours, and these just happen generally to be conspicuous? For patterns too, it may be that some configurations of stimuli (such as the stripes of lepidopteran larvae, or the spotted patterns of coccinelids) are more easily learnt, perhaps because they are 'simpler'. Again, nothing is known about the potential role of memorable patterning in the design of sexual or assessment signals.

Novelty. The effect of novelty on memorability has been well documented by psychologists (e.g. Shettleworth 1972; Domjan 1980; Tarpy 1982; Mackintosh 1983). At least with simple stimuli familiarity retards associative learning ('latent inhibition'), perhaps because a familiar stimulus has already been classified as irrelevant (Channell & Hall 1981). Since dull colours tend to be the commonest in the environment, conspicuous ones may provide unpalatable prey with signals not dogged by the effects of latent inhibition. But conspicuous colours are also novel in the sense that they are unlikely to have conflicting prior associations with palatable prey (which must usually be dull to be

cryptic). Aversion learning does not therefore have to be preceded, and hence slowed, by extinction of previous positive associations (these arguments are further developed in Guilford 1990). Here the properties of memorability and discriminability interact.

Novelty may be important in sexual displays too. If such displays function, at least in part, to enhance the female's associations with a particular male, then we might expect considerable variation among males, each trying to be novel. Novelty does not seem to be a feature of all sexual displays; indeed, some courtship is highly conservative (e.g. Wiley 1973). In such cases, if there is a memorability effect, it is more likely to be due to a male's need to be quickly and reliably recognized (or remembered) as belonging to a particular category, such as 'male' or 'male of the right species', and might work by enhancing the female's generalization to experiences of previous males she has encountered (see Halliday 1991 for other possibilities). On the other hand, perhaps males are anxious to stick out from other events in a female's day, rather than necessarily stick out from other males, so that even relatively conformist displays may be novel in a wider context. Novelty is so dependent on context and experience that its role in many natural displays may easily be overlooked. Nevertheless, it does seem to be at a premium in some displays, such as the highly variable songs of mimics such as starlings, *Sturnus vulgaris*, or the extraordinarily large repertoires of song birds such as mistle thrushes, *Turdus viscivorus* (e.g. Hartshorne 1973; Baylis 1982; Kroodsmma 1982). In some bird species, such as the ruff, *Philomachus pugnax* (Höglund & Lundberg 1989) even adult male plumage can be highly polymorphic. Could novelty here function to enhance the memorability of individual males, their qualities and location?

Potentiation displays. The associability of two stimuli can also be affected by the presence of a third, which can either 'overshadow' (block) or 'potentiate' (enhance) the formation of an association (Mackintosh 1974, 1983). Though it is virtually unstudied, the idea that prey animals might enhance aversion learning in predators by the use of novel or intense potentiating stimuli that do not themselves form part of the warning cue seems plausible. If rats, *Rattus norvegicus*, are sickened

after presentation of a novel taste in conjunction with an odour, they show greater subsequent aversion to the odour than if it is presented on its own during learning (e.g. Rusiniak et al. 1979; Palmerino et al. 1980), whilst in several bird species novel tastes can potentiate associations between colour and unpalatability (e.g. Brett et al. 1976; Clarke et al. 1979; Lett 1980). Distinctive pyrazine odours, which are common in aposematic insects and seem to be emitted during defensive displays (Rothschild 1964b; Rothschild et al. 1984), whilst not apparently unpalatable to birds (Guilford et al. 1987), might serve to potentiate avoidance learning of warning colours (Guilford 1987). Certainly, they can potentiate location-based learning in rats (Kaye et al. 1989), as can tastes (Westbrook & Brooks 1988). Design for memorability in warning signals may therefore extend beyond the characteristics of the warning cues themselves to exploit complex psychological reactions involving the interactions of many different kinds of stimuli at once.

One intriguing possibility is that some sexual signals are 'potentiating displays' designed to enhance the female's association between a male's quality and his identity or location. Displays such as those of birds of paradise are striking in that they use a variety of components in different modalities all at once (sounds, colours, movement; e.g. Gilliard 1969; Frith 1981). It seems more plausible to think of sound potentiating a learnt association in the female's mind than it is to explain such complex displays purely in terms of quality advertisement. If the function is to cause an event to stick out in the female's memory, then perhaps a complex display in several modalities simultaneously would be most impressive. Quality advertisement, on the other hand, would seem to demand a single clear message, not the confusion of a multi-modal signal. Flowers benefit from insects or other animals visiting flowers that look like them. They use nectar as reinforcement (e.g. Pyke 1981), and it is possible that their scent or the colours of their petals are designed to potentiate the learnt association between nectar yield and flower position, time of availability, or visual characteristics. If potentiating displays are involved, the empirical prediction is clear: they should have the curious property of being important in the formation of preferences for, say, particular males or flower types, but unimportant as cues for the subsequent recognition of those males or flower types. Potentiating displays

contain no strategic message at all: they are pure efficacy.

EFFICACY, HONESTY AND MANIPULATION

We have seen that there are many different ways in which receiver psychology can affect the evolution of signal design. Different receivers will have different sense organs and different brains and they will have been selected to respond to different aspects of signal discriminability and detectability. In some cases, memorability will be an additional important design feature, in other cases it will not. Such diversity in what makes a signal effective is precisely what we need to explain the diversity of signals themselves. Another source of diversity, however, is diversity of the messages carried by those signals and we conclude by discussing how the ideas presented here fit in with current ideas of honesty (Zahavi 1987) and manipulation (Krebs & Dawkins 1984) in animal signalling.

Zahavi (1975, 1987) suggested that all signals have been designed reliably to show off a signaller's underlying quality to receivers with respect to some attribute in which the receiver has an interest, in such a way as to benefit both parties. This is the theory of 'honest signalling' which has been gaining increasing acceptance (e.g. Hamilton & Zuk 1982; Kodric-Brown & Brown 1984; Burk 1988; Dawkins 1989; Michod & Hasson 1990) and is regarded by some as a general theory of signal selection (Zahavi 1987; Grafen 1990b). The logic is simply that since it will not pay receivers to respond inappropriately to signals, dishonest signals will become ignored and only those that genuinely reveal quality will produce the responses sought by the signaller. Brightly coloured plumage is seen as being designed to expose parasite load (Hamilton & Zuk 1982; Read 1987) and roaring in red deer to expose battle fitness (Clutton-Brock & Albon 1979). Zahavi (1987) argued that it is a signal's cost that guarantees its honesty, specifically by tapping, and hence testing, the very reservoir of quality that the signal is attempting to advertise: an argument supported by Grafen's evolutionarily stable strategy models of the strategic choice handicap (Grafen 1990a, b).

Although honesty may be an important strategic element in the design of signalling systems, not all signals are necessarily honest (Wiley 1983; Adams & Caldwell 1990; Dawkins & Guilford 1991). As

Dawkins & Krebs (1984) made clear, dishonest signals may also be stable if they elicit responses inappropriate to the receiver but occur rarely enough not to constitute a significant selective force. The colour patterns of Batesian mimics (Edmunds 1974), or the mimetic gape of the cuckoo chick are examples of how signals have evolved to cheat receivers with dishonest messages. Even within a single species, dishonest signals can be stable if they are kept in balance with honest signals by frequency-dependent probing, or if they are useful only to a subset of individuals who are necessarily rare for some other reason (Dawkins & Guilford 1991). The bluffing threat displays of moulting (and therefore vulnerable) stomatopods, *Gonadodactylus bredini*, are constrained to be rare by the timing and duration of moulting itself (Adams & Caldwell 1990). Such signals were called 'manipulative' by Dawkins & Krebs (1978; Krebs & Dawkins 1984).

Krebs & Dawkins (1984) also introduced another concept in communication, that of 'mind-reading'. Receivers can predict the future behaviour of emitters because the emitter unwittingly gives away its intentions. The receiver therefore can exploit the emitter, because it cannot help giving away what it is about to do next. The exploitation of golden plovers, *Pluvialis apricaria*, by gulls that predict when a plover has found a worm (Thompson 1983) is an example of mind-reading. Communication through mind-reading does not involve the special evolution of traits designed to carry information or elicit reactions in the receiver. In fact, it involves the very opposite, with the movements of the mind-read individual being designed, as far as possible, not to evoke a response in the receiver.

Honesty, manipulation and mind-reading, then, can all be seen as strategic components of signal design or, in the case of mind-reading, as a strategic component of non-signal design. Only the receiver benefits in mind-reading, only the emitter in manipulation but both benefit in honesty. In each case, however, the tactical or efficacy component operating through receiver psychology will also be important. With honest signals, the emitter will be selected to emphasize those aspects of a signal that are important to a receiver and easy for it to detect, discriminate or remember. With manipulative signals, the evolutionary arms race between signaller and receiver will involve the signaller making use of receiver psychology for its own ends, followed by selective pressures on receivers not to

respond (i.e. to change their psychology). With mind-reading, what a receiver finds detectable, discriminable or memorable must be avoided and so also has a profound effect on the appearance and behaviour of the 'mind-read' individual. Thus, palatable prey avoid communicating their location to predators by using cryptic patterns that present a random sample of background elements and are therefore difficult for predators to detect (Endler 1978, 1984, 1986). But predators have evolved special mechanisms for learning the characteristics of frequently encountered cryptic prey, or for increasing their detectability by attending selectively to them (such as search image formation), so prey may make use of rare patterns (hence polymorphism) to foil even quite sophisticated aspects of predator psychology (Guilford & Dawkins 1987; Guilford, in press). The strategic component of signal design varies under these different circumstances but in all cases the final appearance of the signal cannot be understood without also taking into account the tactical or efficacy component. Some authors ignore signal efficacy (e.g. Grafen 1990b), whilst others seem to confuse the two. Krebs & Dawkins (1984), for example, regarded signals as 'manipulative' not just if they act to the emitter's but not the receiver's benefit (strategic component), but also if they key into some pre-existing bias in the receiver's responsiveness (tactical component, or efficacy). Yet both are independently important components of signal design. However much we understand about the strategic component of signal design, we will never explain why signals are the way they are and why they differ so greatly from species to species until we have a clearer idea of how they have their effects. Animals are 'Nature's psychologists' (Humphrey 1976). We must be too.

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